

Customers' Purchasing Pattern (CPP) Analysis

A portfolio project applying data analytic methodology to understand customer purchase behavior and generate actionable business insights for enterprise improvement.



What Is Customers' Purchasing Pattern?

Purchasing decisions are rarely spontaneous – they reflect habits shaped by personal, psychological, and social traits. CPP captures these behaviors as a measurable model covering:

How

Customers purchase items

Amount spent

Monetary value of items purchased

Frequency

How often they buy

Recency

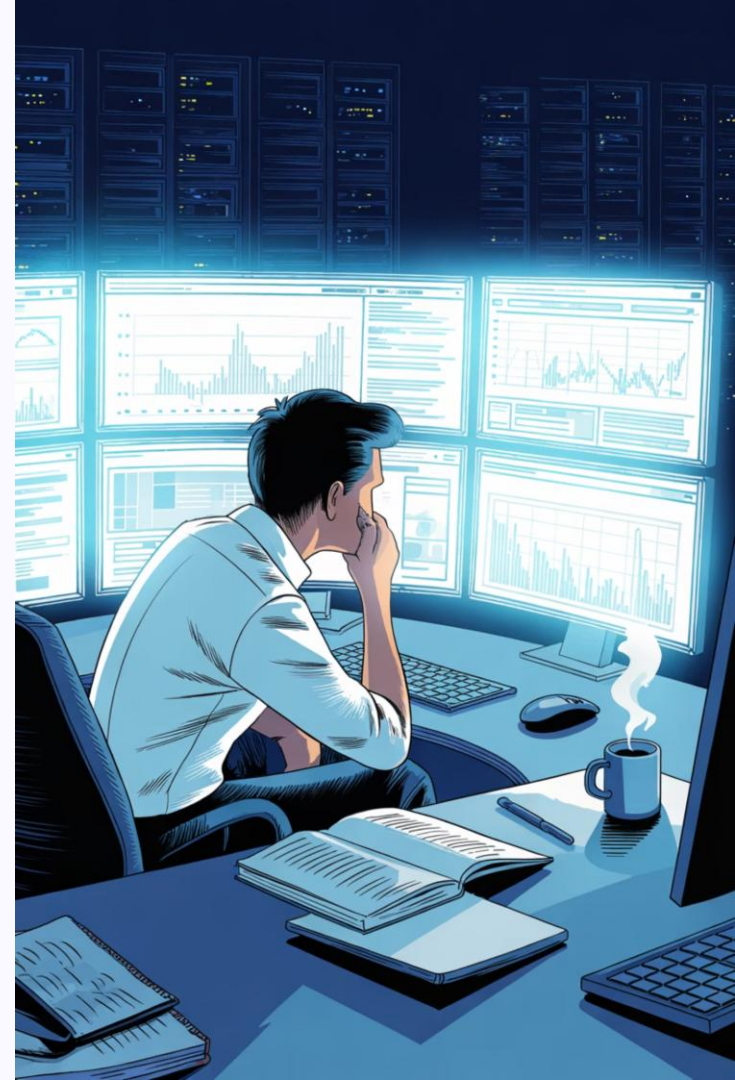
Time interval between purchases

Preferences

What products they favor

Strategy

Marketing & engagement planning



Goal & Objectives of the Portfolio Study

The overarching goal is to demonstrate data analytic methodology applied to customer purchase patterns and generate insights to impact improvement on the enterprise's sales outcomes.

01

Research Questions

Define questions guiding the study and business solutions

02

Real-Life Dataset

Clean and pre-process a secondary online dataset

03

Python Implementation

Write code to solve real-life business issues

04

Generate Insights

Produce recommendations for C-suite and operational staff

05

Document Findings

Report for technocrats and slides for top management

Five Core Business Questions

1

Purchase Drivers

What drives item purchases from the company?

2

Major Customers

Who are the primary purchasers?

3

Purchase Value

What is the monetary value of items purchased?

4

Purchase Frequency

How often does an average customer purchase?

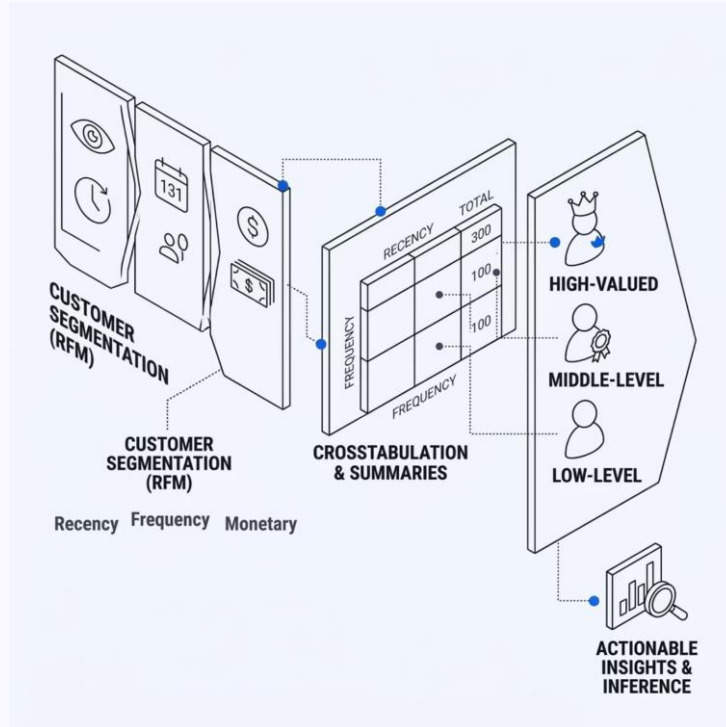
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Recency

How recently (time interval) did an average customer purchase?

The Analytical Framework

A broad-based, integrated 2-level framework was adopted to deliver incisive, actionable insights across social, economic, demographic, and psychographic dimensions.

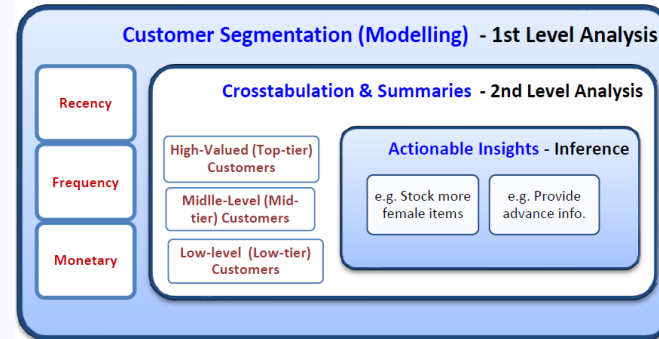


Level 1: RFM Segmentation

Models customers by **Recency**, **Frequency**, and **Monetary** value to create segments.

Level 2: Crosstabulation & Summaries

Classifies customers into **Top-tier**, **Mid-tier**, and **Low-tier** groups, generating actionable insights such as stock recommendations and advance info delivery.



Dataset Properties



Initial Dataset

1,966 instances (rows) and **50 features** (columns) covering purchases and related variables.

Feature Engineering

Created **16 new features** through feature creation, then selected **28 features** best suited for CPP analysis.

Tools

Python in Jupyter IDE for all analytical procedures.

Data Pre-processing Steps

Multiple cleaning and transformation tasks prepared the data for robust CPP analysis.

→ **Label Formatting**

Converted column headers to snake_case format

→ **Missing & Duplicate Handling**

Removed blank data-points and duplicated records

→ **Outlier Detection**

Identified and treated statistical outliers

→ **Feature Creation**

Engineered new columns – e.g., `age_group` via `pd.qcut`

→ **Categorical Encoding**

Encoded categorical variables for modeling

Python Code Highlights

```
# Snake-case column formatting
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')

# Create age_group feature
labels = ['Adolescent', 'Youth',
          'Young Adult', 'Adult']
df['age_group'] = pd.qcut(
    df['age'], q=4, labels=labels)
```

Final Dataset Shape

1,966 entries, 31 columns after feature selection and engineering.

Key Variables

- customer_no, recency, order_freq
- customer_segment, yearly_income
- age, sex, age_group
- dir_spent_multi_purch

Customer Segmentation Results

Three segments were extracted from the MRF dataset for targeted analysis.

1,966

Total Records

Full dataset instances analyzed

109

Premium Customers

Segment 3 – Highest-value buyers

933

Opportunity Customers

Segment 2 – Mid-tier with growth potential

924

Least – Buying Customers

Segment 1 – Lowest engagement tier



Key Takeaways

Robust Framework

2-level RFM + Crosstabulation approach enables deep, actionable segmentation

Clean Data Foundation

Rigorous pre-processing of 1,966 records across 28 engineered features ensures reliability

Segmented Insights

Three distinct customer tiers – Premium (109), Opportunity (933), Least (924) – enable targeted strategy

Business Impact

Findings support C-suite and operational decisions on marketing, stock, and engagement

CPP Analysis:

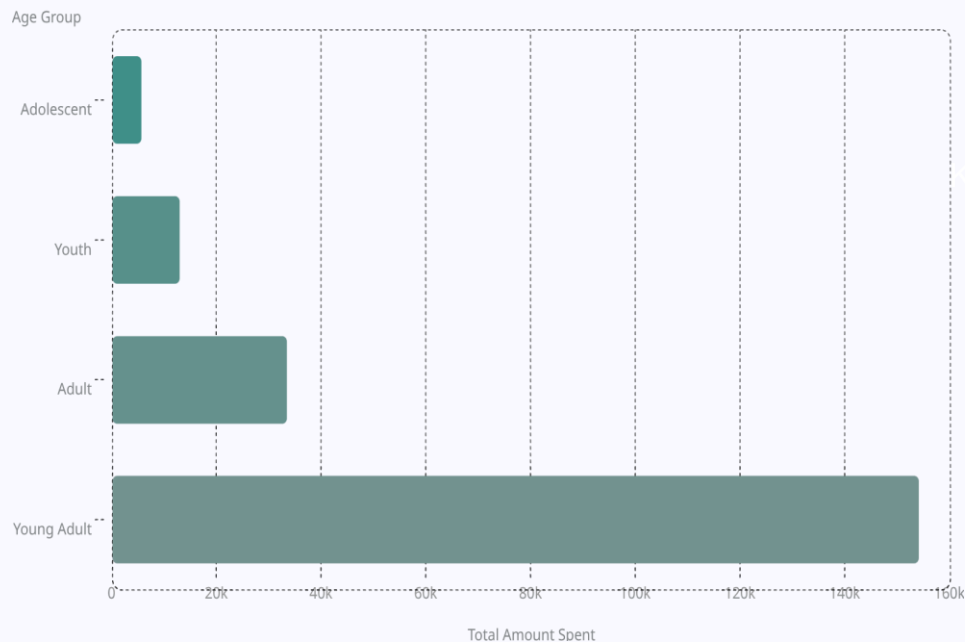
Exploratory Data Analysis (EDA) of MFR Dataset

Exploratory Data Analysis examine - **three customer segments** — Top-tier (Premium), Mid-tier (Opportunity), and Low-tier (Low-buying) — across age, age-group, and sex dimensions.

Results are presented in tables, bar charts, pie charts, and bivariate MFR scatterplot analysis.



Previous Purchases: Spending by Age Group



Key Finding

Young Adults (25–35) spent **significantly more than all other age groups combined**, followed by Adults (36–65). Adolescents (10–15) spent the least.

Average spending also highest for Young Adults (\$2,053) and Adolescents (\$1,838), with Adults and Youths nearly equal (~\$1,514 and ~\$1,425).

- ❶ Implement loyalty rewards for Young Adults; deploy sales promotions to grow Adults, Youths, and Adolescents.

TOP-TIER · PREMIUM CUSTOMERS

Previous Purchases: Spending by Sex

Key Finding

Female Dominance

Females account for **72.8%** of total spending (\$149,709 vs \$55,872 male). Average spending nearly equal: Female \$1,895 vs Male \$1,852.

Current Purchase Trend

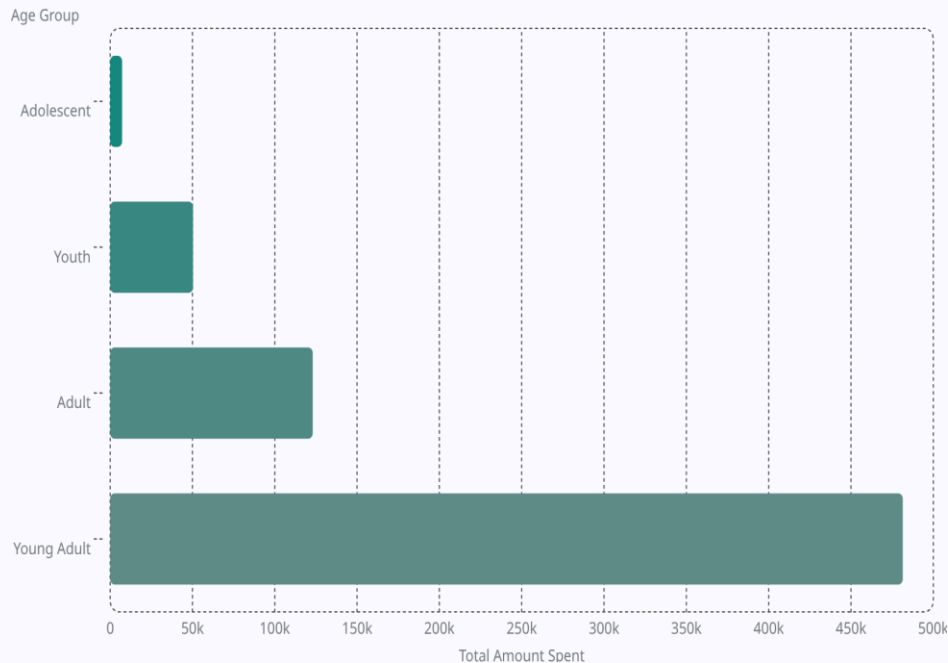
Pattern holds: Females 70.8% of total current spend (\$3,388 vs \$1,404). Average spend also nearly equal: Female \$42.63 vs Male \$46.80.

Strategic Actions

Reward female loyalty to sustain patronage. Launch targeted campaigns to grow the male customer base.



Previous Purchases: Spending by Age Group



Key Finding

Young Adults spent **far more than all other age groups combined**, followed by Adults. Average spending nearly equal across all groups (~\$656–\$720), with Young Adults marginally highest.

Current purchases mirror this: Young Adults led total spend (\$22,299), while Youths had the highest average (\$35.36).

i Reward Young Adult loyalty; deploy promotions to grow Adults, Youths, and Adolescents.



MID-TIER · OPPORTUNITY CUSTOMERS

Previous Purchases: Spending by Sex

Female Dominance

Females account for **80.6%** of total spending (\$532,926 vs \$128,617 male).
Average spending: Female \$727.55 vs Male \$640.48 — marginally higher.

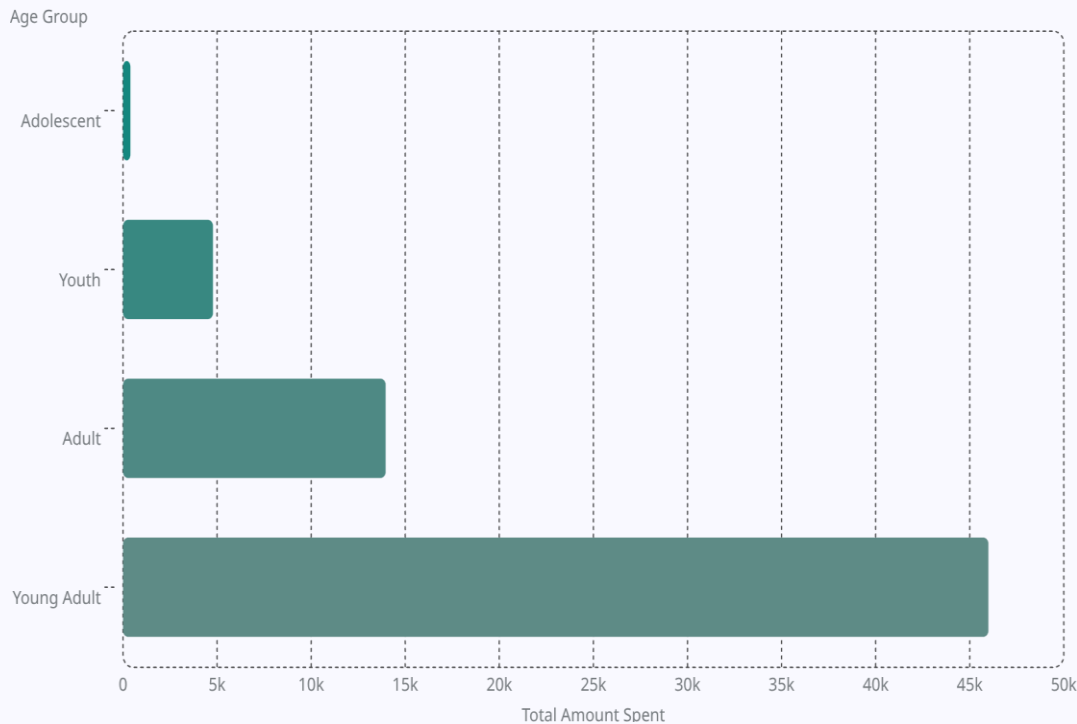
Current Purchase Trend

Pattern holds: Females 79.6% of total current spend (\$24,556 vs \$6,301).
Average spend: Female \$33.55 vs Male \$31.55.

Strategic Actions

Reward female loyalty. Launch male acquisition campaigns. Encourage all Mid-tier customers to step up to Top-tier.

Previous Purchases: Spending by Age Group



Key Finding

Young Adults spent **significantly more than all others combined**, followed by Adults. Average spending: Adults (\$72.70) and Young Adults (\$71.86) nearly equal; Adolescents lowest (\$39.10).

Current purchases follow the same pattern: Young Adults led total spend (\$17,605), with Youths and Young Adults highest on average.

📌 Prioritize Adolescent engagement with age-relevant promotions. Reward Young Adult loyalty to maintain patterns.



LOW-TIER · LOW-BUYING CUSTOMERS

Previous Purchases: Spending by Sex

Female Dominance

Females account for **67.3%** of total spending. Average spending nearly equal: Female 51.2% vs Male 48.8%.

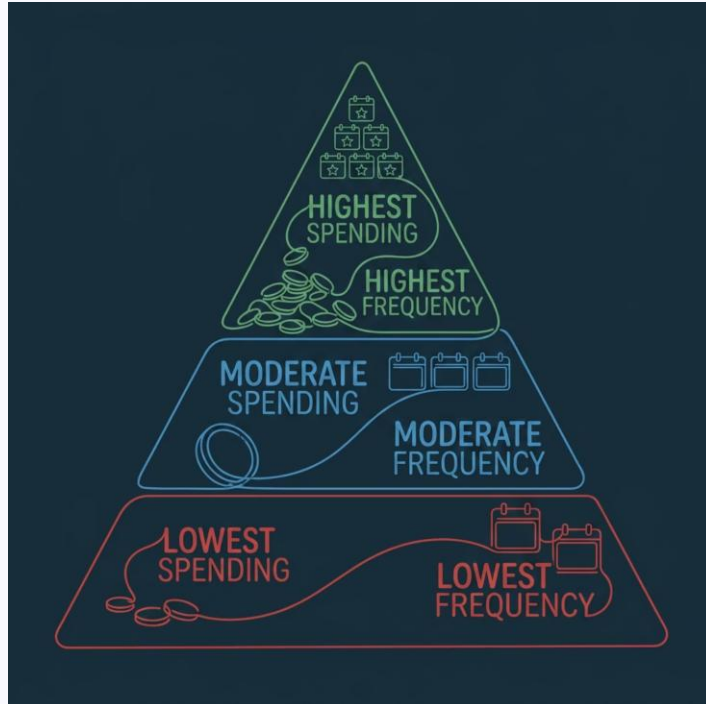
Current Purchase Trend

Pattern holds: Females 66.7% of total current spend. Average spend: Female 56.2% vs Male 43.8%.

Strategic Actions

Reward female loyalty. Launch male acquisition campaigns.
Encourage all Low-tier customers to step up to Mid-tier or Top-tier.

Spending vs Order Frequency Across Segments



Segment Insights

Top-tier:

- Highest spend and frequency. Stock their most-purchased items; promote less-bought products to increase basket size.

Mid-tier:

- Moderate spend and frequency. Deploy promotions to boost both metrics and encourage transition to Top-tier.

Low-tier:

- Lowest spend and frequency. Conduct satisfaction surveys to identify barriers; apply targeted interventions.

Spending vs Purchase Recency Across Segments



Segment Insights

Top-tier:

- Highest spend and most recent orders. Sustain strategies that satisfy them; solicit feedback to further boost spend and shorten order intervals.

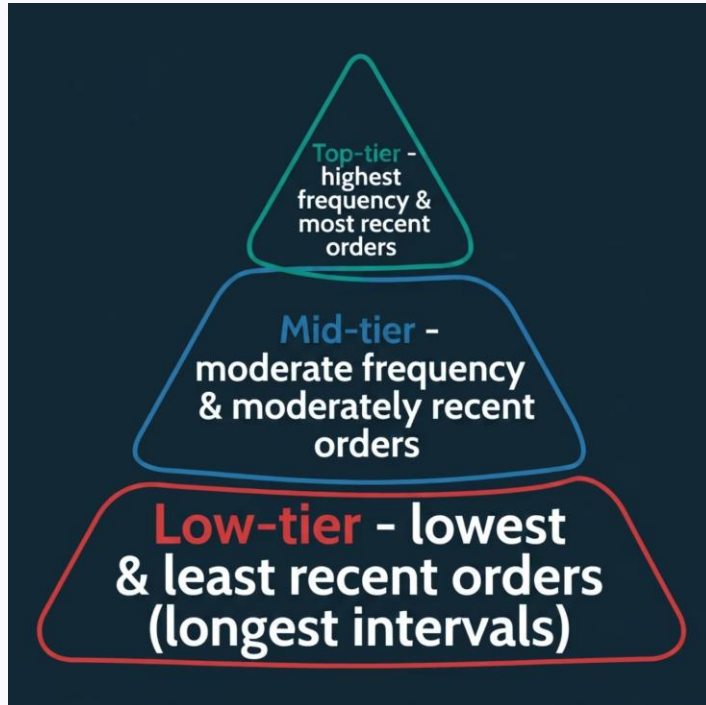
Mid-tier:

- Moderate spend and recency. Target with promotions to increase spend and shorten order intervals.

Low-tier:

- Lowest spend and least recent orders. Conduct satisfaction surveys; apply targeted interventions to boost spend and reduce purchase intervals.

Order Frequency vs Recency Across Segments



Segment Insights

Top-tier:

- Most frequent and most recent orders. Implement loyalty rewards to sustain high frequency and short intervals.

Mid-tier:

- Moderate frequency and recency. Target with promotions to increase frequency and shorten order intervals.

Low-tier:

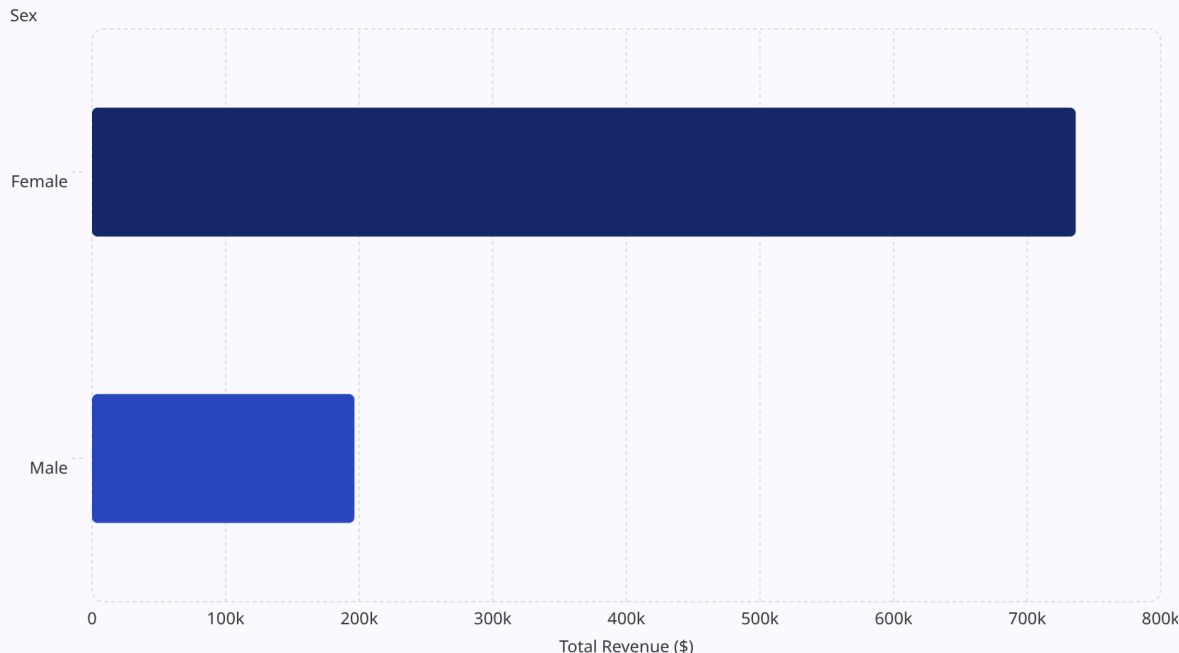
- Least frequent and least recent orders. Conduct satisfaction surveys to understand low patronage; apply targeted interventions to increase frequency and reduce purchase intervals.

CPP Business Intelligence Insights

SQL- driven analysis of customer purchasing patterns — segmented by sex, age group, subscription status, and customer tier — to inform strategic sales planning.



Revenue by Customer Sex



Key Insight

Female customers generated **\$736,354** in revenue — nearly **4× more** than male customers (\$196,462). This represents a significant purchasing gap worth addressing in targeted marketing strategies.



High-Value Customer Segments

576

High Income + High Spenders

Customers earning >\$100K annually
with above-average total spending

249

Frequent Buyers

Customers with 5+ orders and above-
average total spending

575

Recently Active

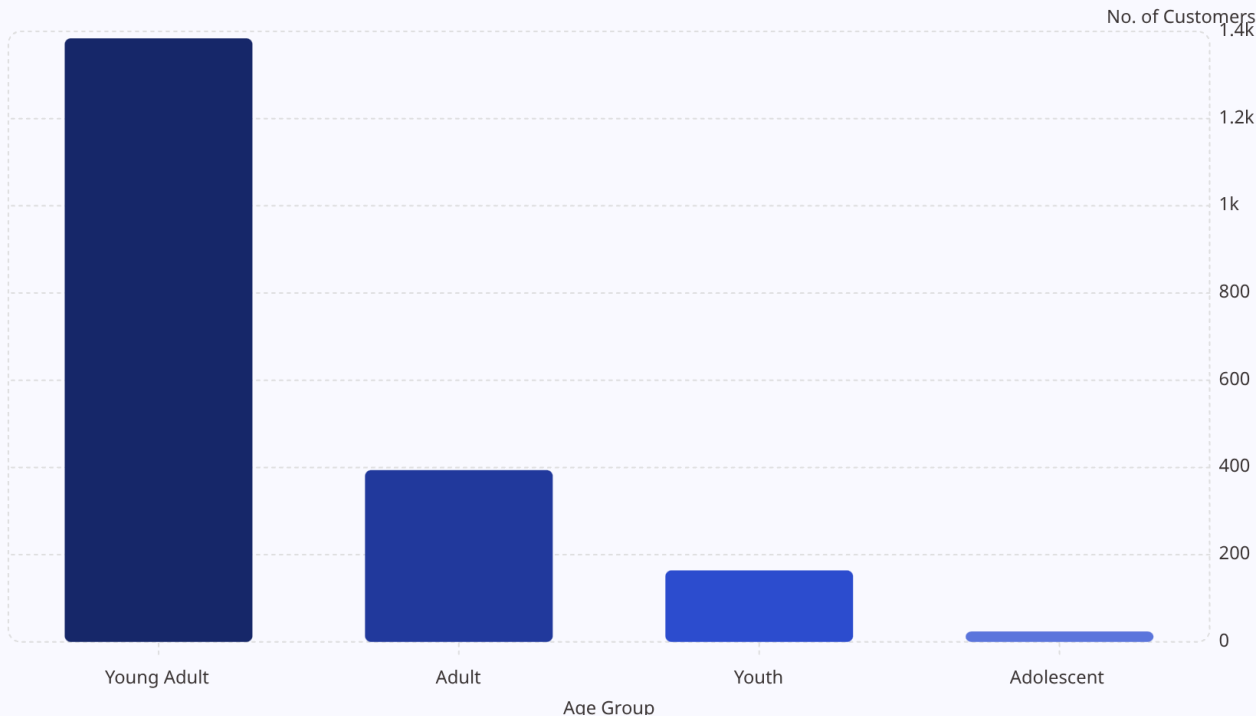
Customers who purchased within 2
years with above-average total spending

Top 10 Recently Purchased Items

Item	Total Amount (\$)	Average Rating
Coat	3,166	3.89
Pants	2,957	3.78
Shorts	2,853	3.83
Sweater	2,528	3.87
Sneakers	2,409	3.90
Belt	2,357	3.84
Skirt	2,322	3.83
Blouse	2,201	3.78
Socks	1,945	3.77
Dress	1,872	3.98

Dress leads in average rating (3.98) despite lower sales volume. **Coat** generates the highest total revenue among recent purchases.

Customers by Age Group

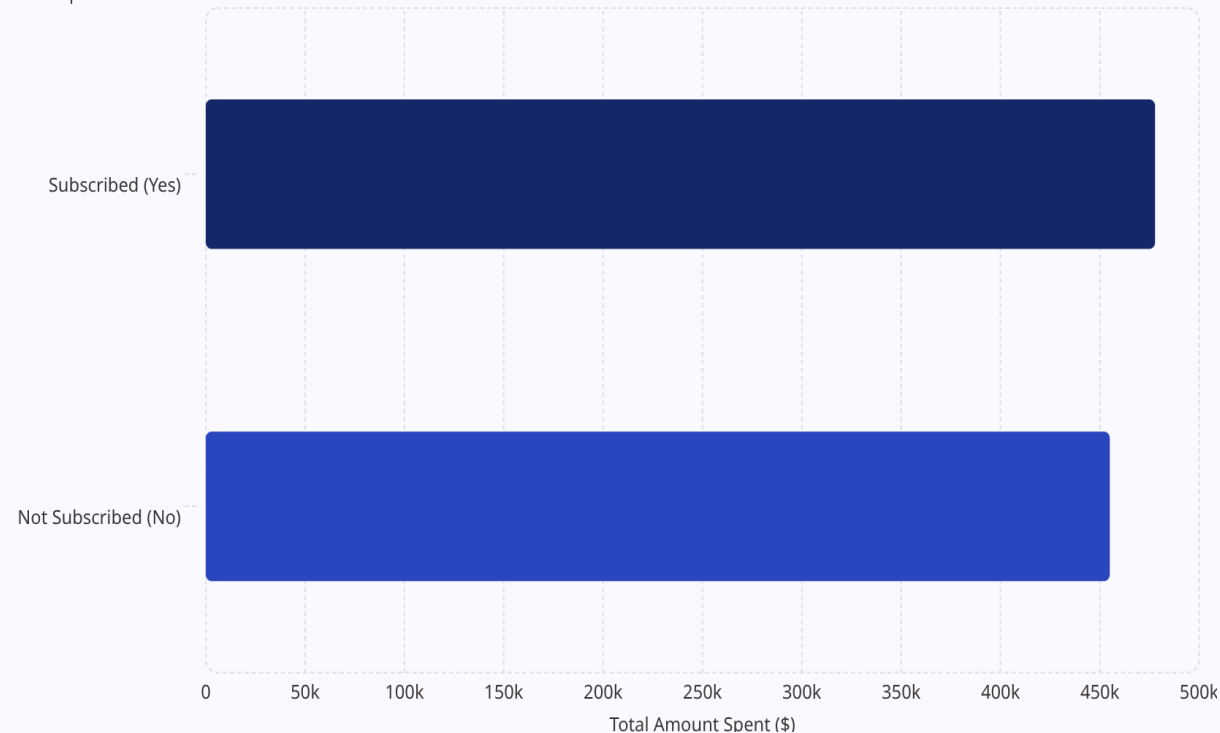


Spending by Age Group

Young Adults dominate in volume (1,384 customers) and total spend (\$681,367 historically, \$43,120 recently). However, **Adolescents** spend the most per customer on average (\$546.88), despite being the smallest segment.

Subscription Status Analysis

Subscription



Subscribed: 973 customers

Avg spend: **\$491.05** |

Total: **\$477,794**

Not Subscribed: 993 customers

Avg spend: **\$458.23** |

Total: **\$455,022**

Subscribers spend **\$33 more on average** per customer, despite slightly fewer total subscribers.



Customer Segments

Low-Tier

924 customers —
largest segment

Mid-Tier

933 customers — nearly
equal to Low-Tier

Top-Tier

109 customers — premium segment

Top 3 Items by Customer Segment

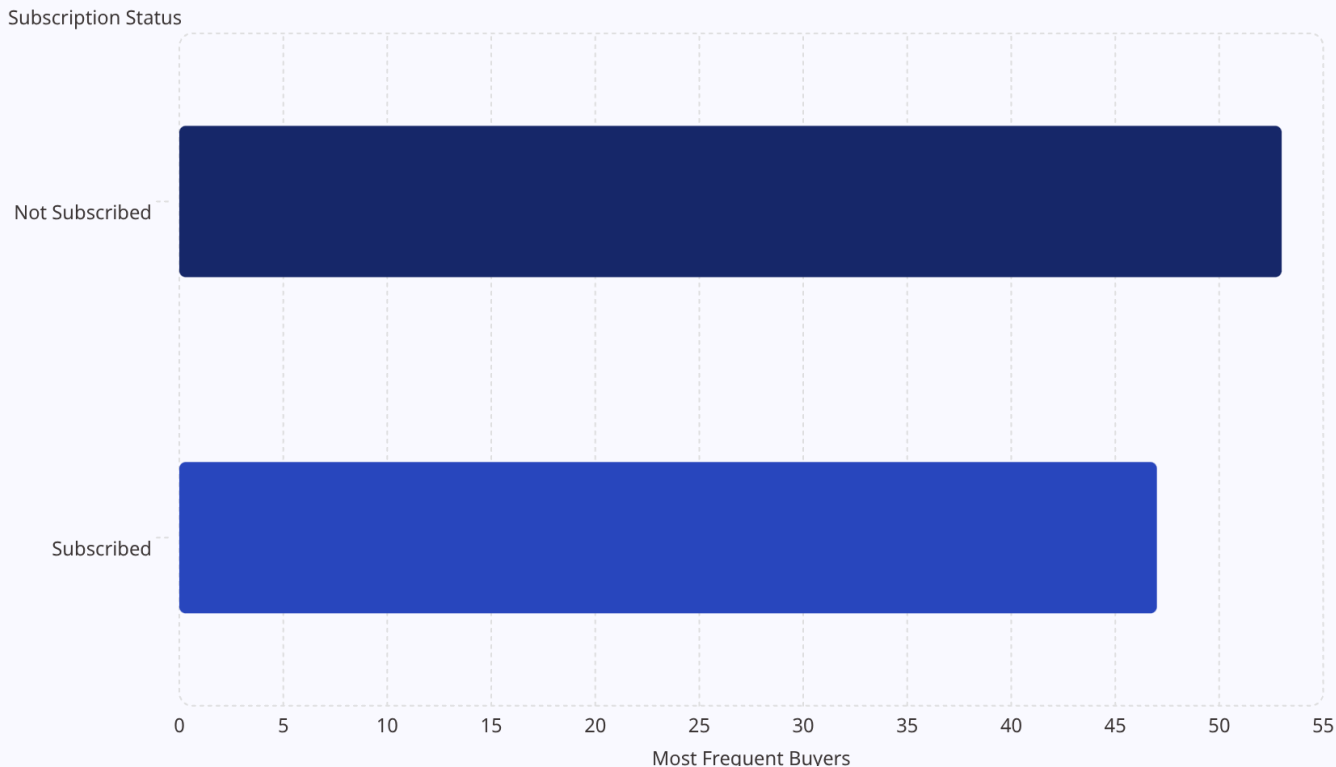
Segment	Rank	Item	Total Amount (\$)
Low-Tier	1	Coat	1,872
Low-Tier	2	Boots	1,396
Low-Tier	3	Jacket	1,280
Mid-Tier	1	Hoodie	1,631
Mid-Tier	2	Pants	1,621
Mid-Tier	3	Boots	1,617
Top-Tier	1	Shorts	351
Top-Tier	2	Sweater	347
Top-Tier	3	Shoes	335

- * **Boots** appear in both Low-Tier and Mid-Tier top purchases.
- * Top-Tier customers show distinctly different preferences, favoring casual items like **Shorts** and **Sneakers**.

Frequent Buyers by Subscription Status

Surprising Finding

Non-subscribers account for **53** of the most frequent buyers vs. **47** subscribers — suggesting that subscription status alone does not drive purchase frequency. Retention strategies may need to extend beyond subscription incentives.



Key Takeaways & Strategic Priorities

→ Focus on Female Customers

Female customers drive 79% of revenue — prioritize targeted campaigns for this segment.

→ Convert High-Value Non-Subscribers

576 high-income, high-spending customers represent untapped subscription revenue.

→ Upsell to Young Adults

Largest segment by volume — highest recent spend (\$43,120) signals strong growth potential.

→ Re-engage Top-Tier Customers

Only 109 top-tier customers — personalized outreach could boost their low recent purchase totals.



CPP Analysis Dashboard

Business Intelligence Dashboard —
Customer Purchasing Patterns, Segmentation & Revenue Insights

Customer Overview

1.97K

Total Customers

79%

Female

vs. 21% Male

993

No Subscription

vs. 973 Subscribed

The customer base is predominantly female (79.45%). Subscription status is nearly evenly split — 993 without and 973 with active subscriptions.

Currently Purchased Items — Key Metrics

999

Active Buyers

Customers who most
recently purchased

\$30.65

Avg. Spend

Per customer on most
recent purchases

\$60.26K

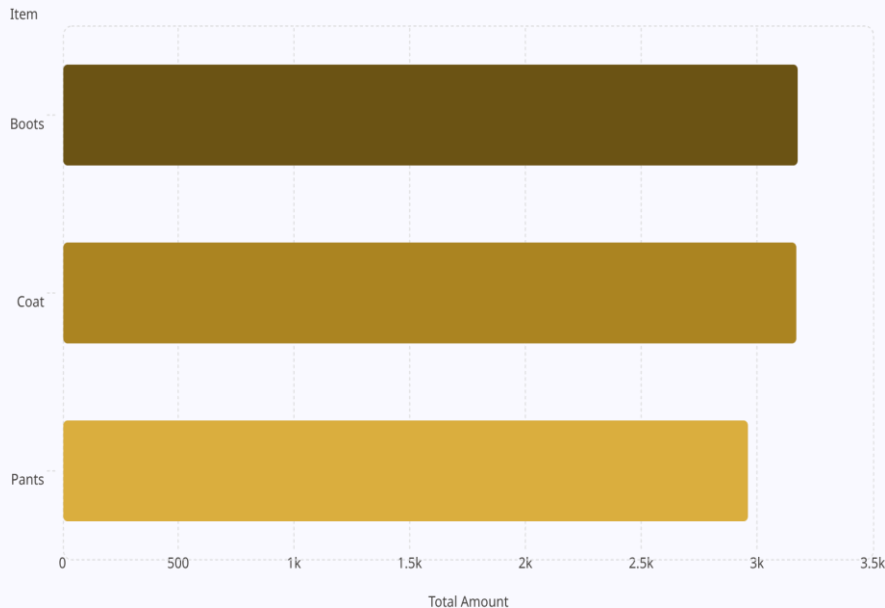
Total Revenue

From current item purchases

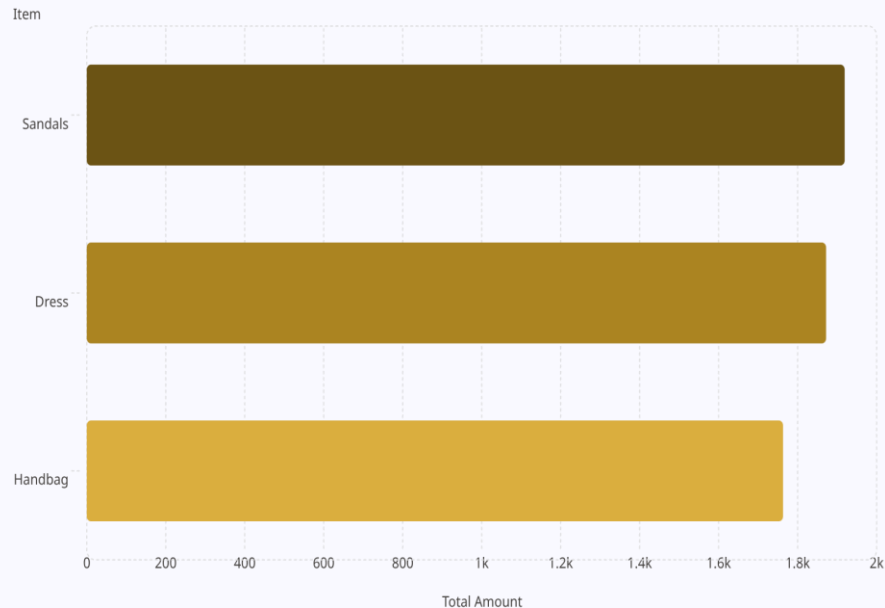


Top & Bottom Items — Most Recent Purchases

🏆 Top 3 Items by Total Amount



📉 Bottom 3 Items by Total Amount



- Boots, Coat, and Pants lead recent purchase volume, while
- Sandals, Dress, and Handbag rank lowest among all items tracked.

All-Time Previous Purchase Metrics

Average Amount Spent

\$474.47 per customer on previous item purchases

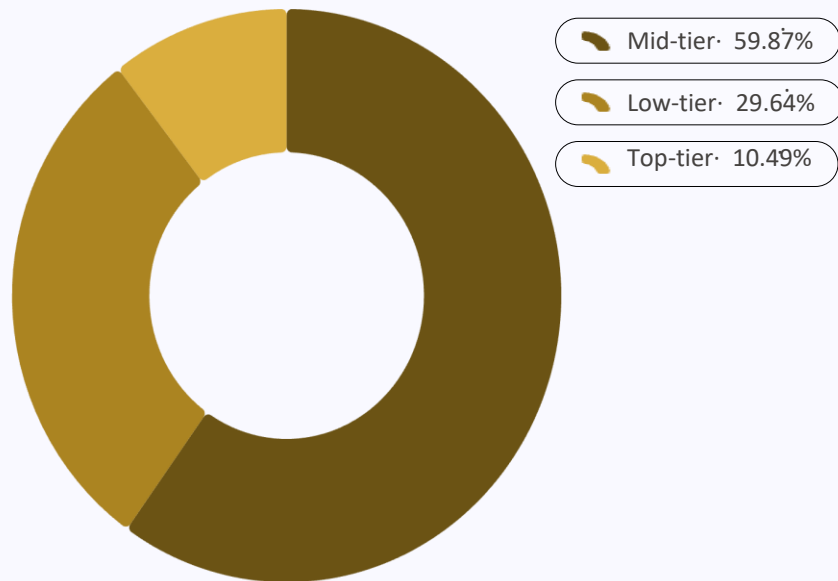
Total Amount Spent

\$932.82K cumulative revenue from all-time previous purchases



Customer Segment Breakdown — MRF Analysis

Segment Distribution

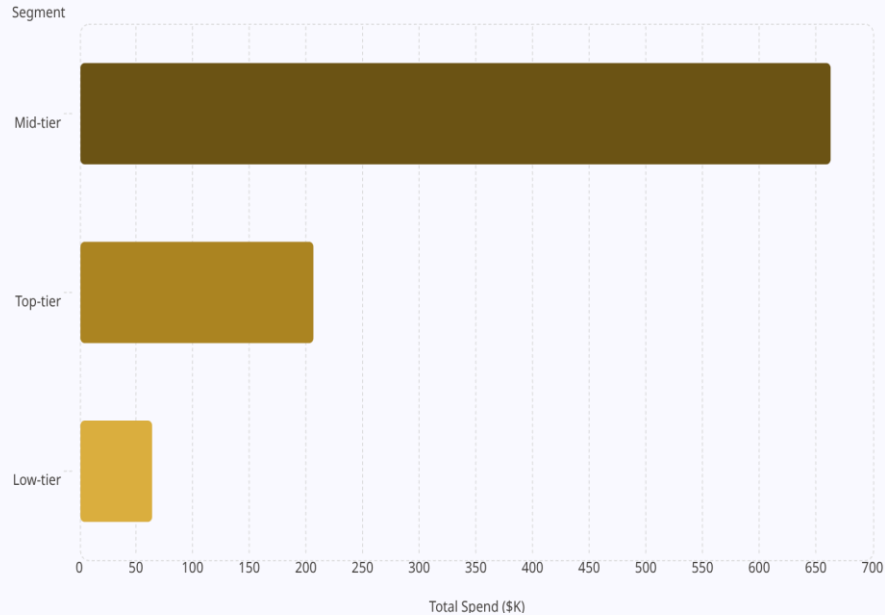


Segment Definitions

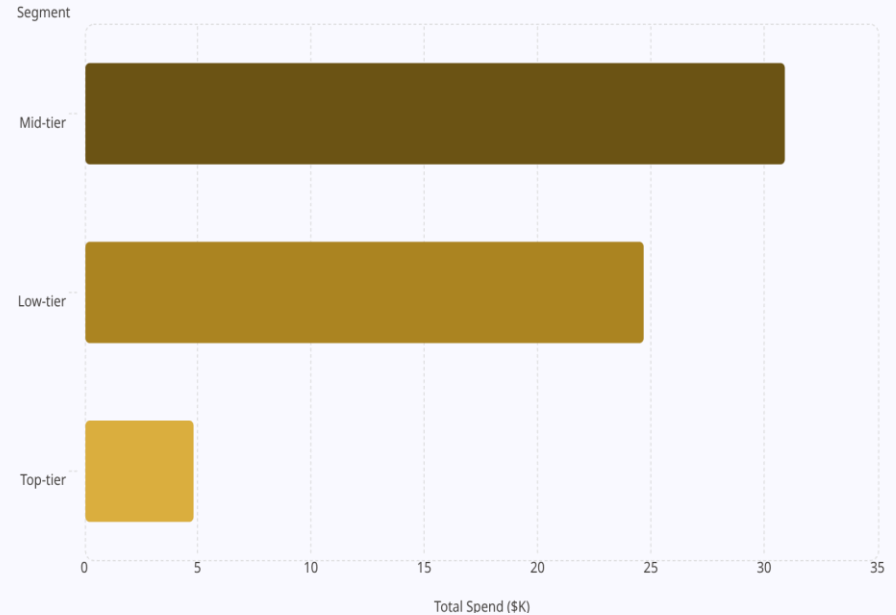
- **Mid-tier (59.87%)** — Largest segment; opportunity customers with moderate purchasing frequency
- **Low-tier (29.64%)** — Least-buying customers; lowest engagement and spend
- **Top-tier (10.49%)** — Premium customers; highest value despite smallest share

Revenue by Customer Segment

Previous Purchases — Total Spend



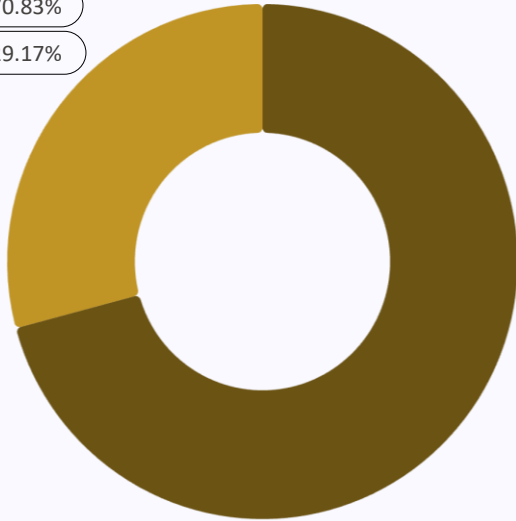
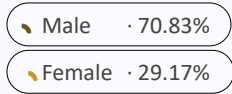
Recent Purchases — Total Spend



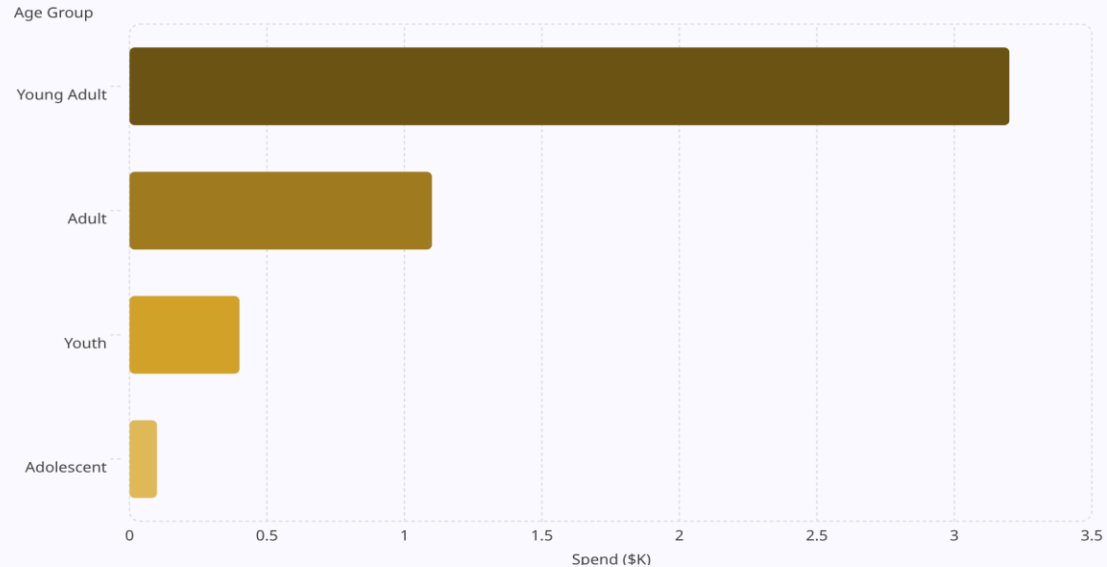
- Mid-tier dominates both previous (\$661.99K) and recent (\$30.86K) spending.
- Top-tier leads in historical value but shows lower recent activity.

Top-Tier Customer Profile

By Sex



By Age Group — Recent Spend



1. Top-tier is **70.6%** male.
2. Young Adults drive the most recent spend (**\$3.2K**).
3. Top items: **Shorts** (351), **Sweater** (347), **Shoes** (335).
4. Subscription: **56** No / **53** Yes.

Mid-Tier & Low-Tier Customer Profiles

MID-TIER — OPPORTUNITY

- a. **79.6%** Male.
- b. Young Adults lead spend (**\$22.3K**).
- c. Top items: **Hoodie** (1,631), **Pants** (1,621), **Boots** (1,617).
- d. Subscription: **474** Yes / **459** No.

LOW-TIER — LEAST-BUYING

- a. **84.7%** male.
- b. Young Adults lead spend (**\$17.6K**).
- c. Top items: **Coat** (1,872), **Boots** (1,396), **Jacket** (1,280).
- d. Subscription: **478** No / **446** Yes.

Across all segments, males dominates, and Young Adults are the primary spenders. Low-tier customers show the weakest recent engagement despite a comparable subscription split.

Conclusion & Strategic Outlook

CPP Defined

Customer Purchasing Patterns are shaped by personal traits, demographics, and psychographics. The MRF framework effectively identified three distinct customer categories.

Segment Insights

Each segment exhibits unique purchasing behaviors — actionable intelligence for targeted marketing and sales improvement strategies.

Forward Strategy

Businesses must invest in top-notch customer service, MRF knowledge, and long-term benefits that align purchasing patterns with company strategy.

